Descrierea Algoritmului Inteligent

- Ramura Vizuala: EfficientNet (efficientnet_b0) pre-antrenat (timm), configurat ca extractor de caracteristici (num_classes=0).
- Ramura Tabulara: Un Perceptron care proceseaza metadatele pacientilor.
- Abordarea 1 - Fuziune Concatenare: vectorii vizuali si tabulari sunt concatenati direct (torch.cat).
- Abordarea 2 - Fuziune prin Cross Attention: utilizeaza straturi de tip MultiheadAttention

- Sursa de Date: https://www.kaggle.com/datasets/ajithdari/multi-modal-breast-cancer-dataset
- Antrenare si Validare: 80% antrenare, 20% testare. Optimizarea s-a realizat folosind algoritmul Adam.

Rezultate Obtinute

Rezultate - Concatenare Simpla: Age at Diagnosis, Neoplasm Histologic Grade:
--- Epoch 1/3 --- Train Loss: 0.5776 | Test Loss: 0.4771 | Test Accuracy: 75.64%
--- Epoch 2/3 --- Train Loss: 0.2987 | Test Loss: 0.3309 | Test Accuracy: 85.90%
--- Epoch 3/3 --- Train Loss: 0.1259 | Test Loss: 0.2525 | Test Accuracy: 91.03%

Rezultate - Concatenare Simpla: Age at Diagnosis | Tumor Size | Lymph nodes examined positive | Nottingham prognostic index | Mutation Count | Neoplasm Histologic Grade | Tumor Stage | ER Status:
--- Epoch 1/3 --- Train Loss: 0.5424 | Test Loss: 0.4770 | Test Accuracy: 76.28%
--- Epoch 2/3 --- Train Loss: 0.2789 | Test Loss: 0.3461 | Test Accuracy: 85.90%
--- Epoch 3/3 --- Train Loss: 0.1350 | Test Loss: 0.3247 | Test Accuracy: 83.33%

Rezultate - Cross Attention: Age at Diagnosis, Neoplasm Histologic Grade:
--- Epoch 1/3 --- Train Loss: 0.5918 | Test Loss: 0.5285 | Test Accuracy: 75.00%
--- Epoch 2/3 --- Train Loss: 0.4141 | Test Loss: 0.3580 | Test Accuracy: 85.90%
--- Epoch 3/3 --- Train Loss: 0.1641 | Test Loss: 0.2997 | Test Accuracy: 89.74%

Rezultate - Cross Attention: Age at Diagnosis | Tumor Size | Lymph nodes examined positive | Nottingham prognostic index | Mutation Count | Neoplasm Histologic Grade | Tumor Stage | ER Status:
--- Epoch 1/3 --- Train Loss: 0.6084 | Test Loss: 0.5464 | Test Accuracy: 73.08%
--- Epoch 2/3 --- Train Loss: 0.4450 | Test Loss: 0.3451 | Test Accuracy: 88.46%
--- Epoch 3/3 --- Train Loss: 0.1739 | Test Loss: 0.2777 | Test Accuracy: 91.03%